Table 1: Survey of the extreme-temperature YSZ grain-growth configurations attempted in the induction furnace, in chronological order. Configurations list the sample assembly top to bottom as loaded in the chamber. Temperatures are ratio-pyrometer readings and are nominal for the run; durations are time at temperature. Grain sizes are from optical micrographs of the recovered specimens. Trace files name the raw heating-profile logs retained in the laboratory archive; runs without an entry predate systematic trace retention. The 2500 °C / 45 min row is the successful anneal reported in the main text (Fig. 7).

Date	Nominal T	Duration	Configuration (top → bottom)	YSZ grain growth	Outcome	Trace file(s)
2023-07-04	2000 °C	80 h	BN / YSZ / Ta	—	Specimen not usable for grain-size measurement.	`2000forWeek.fig`, `fullTime.xlsx`
2024-01-31	2000 °C	24 h	Ta (20 mm) / YSZ / Ta (20 mm) / alumina rod / Teflon rod	10 µm → 20 µm	Growth too slow at 2000 °C; motivated moving subsequent runs to 2500 °C.	—
2024-03-11	2500 °C	45 min	Ta (20 mm) / YSZ / Ta (20 mm) / BN crucible / alumina rod / Teflon rod	20 µm → 90 µm	First successful 2500 °C run; clean specimen. Quartz tube softened, deformed onto the Ta, and was punctured; the interlock stopped the run. Lesson adopted: fixture the quartz with no-slack pyrometer cables.	—
2024-04-29	2475 °C	8 h	Ta / YSZ / BN	∼10 µm → ∼400 µm (tentative)	Stable for the full 8 h. Ta vapor deposited on the YSZ; the specimen was polished and thermally etched before grains could be measured, so the final size is uncertain. Quartz near the Ta softened slightly.	`Ta on top.xlsx`, `2500for8hr.xlsx`
2024-05-23 –27	2130 °C	80.4 h	Ta (large) / YSZ / Ta (thin) / BN / alumina / Teflon	— (specimen unsalvageable)	Susceptor degraded mid-run (a shell, possibly silica–BN–Ta, formed around the Ta); temperature fell ∼500 °C over ∼5 h. Quartz and BN required replacement.	`23May2024.lvm`, `weekLongHeat.fig`
2024-07-22	1750 °C	168 h	Graphite crucible with YSZ inside, on BN	— (specimen consumed)	YSZ reacted with the graphite and disappeared (carbothermal attack). The pyrometer likely under-read this run (the susceptor sat partially outside its view); the true temperature was probably closer to 2000 °C.	`22July2024.lvm`
2024-08-07/08	>2500 °C	2 h	Graphite crucible	—	Pyrometer misaligned; the rise/drop interlock shut the furnace down. Graphite demanded excessive generator power above 2500 °C, motivating the switch to a large tantalum susceptor block.	—
2024-08	1800 °C	115 h	Ta susceptor (configuration not recorded)	—	Quartz–Ta reaction products formed (“geode” growths; no contact required).	—

Alt text: Table of eight extreme-temperature YSZ runs listing date, nominal temperature from 1750 to over 2500 degrees Celsius, duration from 45 minutes to 168 hours, the stacked sample-assembly configuration, measured grain growth where a specimen survived, the outcome or failure mode, and the retained raw trace files. Only the 45-minute run at 2500 degrees Celsius produced a clean specimen with measured growth from 20 to 90 micrometers; longer runs failed by contact reactions between the susceptor, specimen, and surrounding materials.